Nested Ranges Check

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2168 | Sorting and Searching | https://cses.fi/problemset/task/2168

Given n ranges, your task is to determine for each range if it contains some other range and if some other range contains it.

Range $[a,b]$ contains range $[c,d]$ if $a \le c$ and $d \le b$.

Input

The first input line has an integer n: the number of ranges.

After this, there are n lines that describe the ranges. Each line has two integers x and y: the range is $[x,y]$.

You may assume that no range appears more than once in the input.

Output

First print a line that describes for each range (in the input order) if it contains some other range (1) or not (0).

Then print a line that describes for each range (in the input order) if some other range contains it (1) or not (0).

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x < y \le 10^9$

Example

Input

```
4
1 6
2 4
4 8
3 6
```

Output

```
1 0 0 0
0 1 0 1
```